BlackBox

Privacy-first ML marketplace
powered by zero-knowledge proofs

Rise in

ML models demand trust from both sides. Neither side can afford to give it.

01

Healthcare

Hospitals need private diagnostic models but can't send patient data to untrusted endpoints. Model owners can't expose trained thresholds.

02

Finance

Banks need fraud-detection predictions but face regulatory barriers to sharing customer data. Providers won't open-source scoring weights.

03

HR and recruiting

Organizations want unbiased candidate evaluation without revealing proprietary screening algorithms or candidate personal data.

04

Marketing

Advertisers need personalized targeting but GDPR and CCPA prohibit processing user data in the clear.

Introducing BlackBox

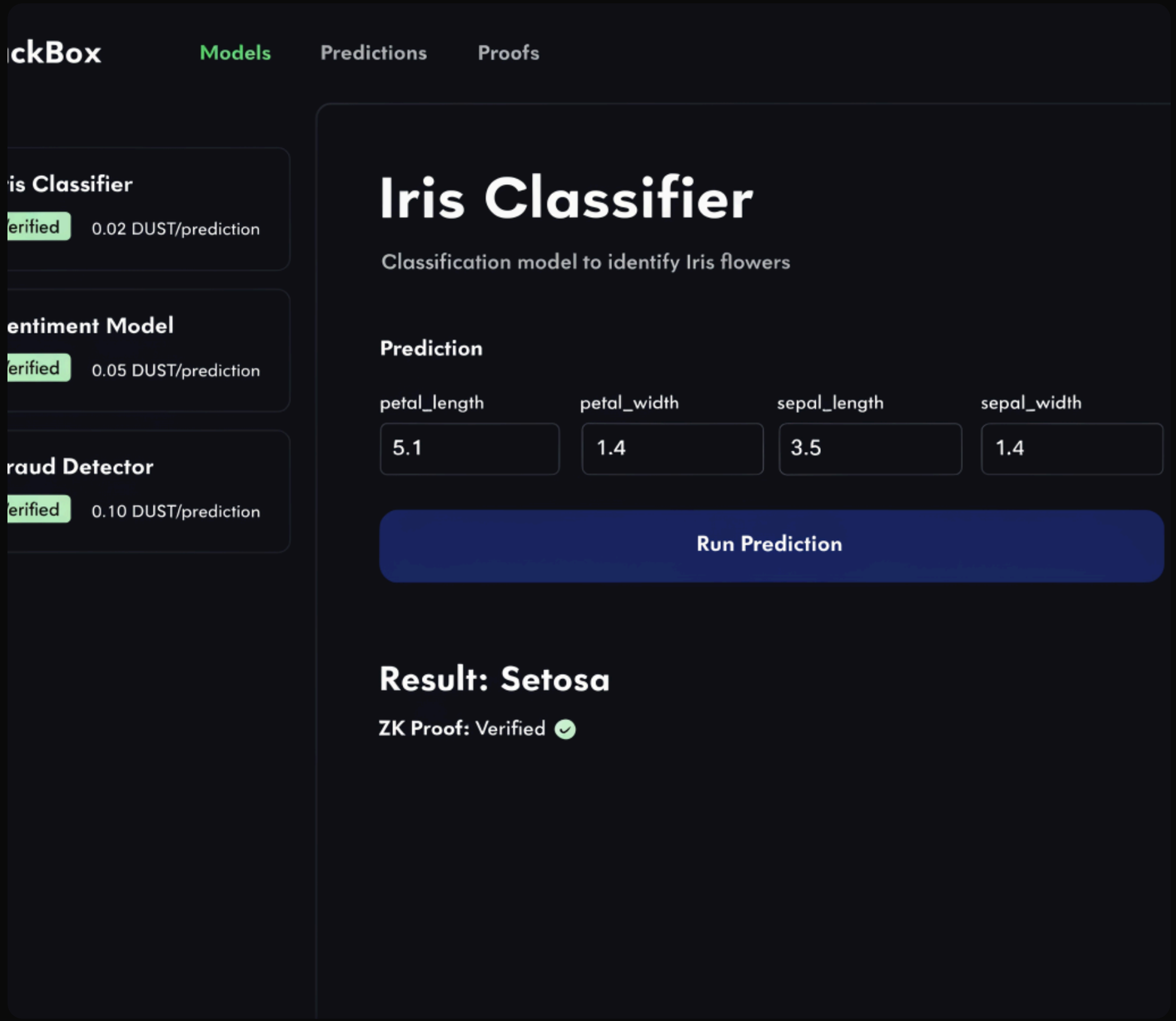
A decentralized pay-per-prediction ML marketplace. No model leaks. No data exposure.

Total model privacy

1. Model logic, thresholds, and training data never leave the provider's infrastructure
2. Every prediction includes a zero-knowledge proof of correct computation

Trustless audit trail

1. Proof verification recorded on Midnight Network blockchain
2. Immutable, privacy-preserving records with selective disclosure



01 Upload model logic

Provider uploads their ML model as a guest binary (ELF). Model thresholds and branching logic remain fully private, never exposed.

02 Request prediction

User sends input data to the RISC Zero host. The model executes inside the zkVM - all computation happens privately, with no access to internal logic.

03 Generate ZK proof

RISC Zero zkVM produces a cryptographic proof of correct execution. Only the final prediction is committed to the proof journal. No model internals leak.

04 Verify on-chain

Midnight's Compact smart contract verifies the proof on-chain. User receives the prediction plus a verifiable record with selective disclosure controls.

Proof of concept Iris classification with verifiable, private inference

Input and output

- 4 floating-point measurements as input
- Sepal length, sepal width, petal length, petal width
- Classifies into 3 Iris species
- Setosa, versicolor, or virginica

Decision tree model

- Hardcoded thresholds for branching logic
- Compiled into a RISC-V guest binary
- Runs the full classification pipeline
- Classic Iris dataset as training source

Zero-knowledge proof

- Entire inference runs inside the zkVM
- Produces cryptographic proof of correctness
- Tree branching logic stays fully hidden
- Verifiable output without revealing the model

Key features
Privacy, verifiability, and control built in

Private inference
Run predictions without exposing model weights or training data. Full confidentiality for providers and consumers.

Pay-per-request
Model providers monetize each inference call. No bulk licensing, no IP handoff. Revenue scales with usage.

Verifiable outputs
Zero-knowledge proofs confirm prediction correctness without revealing model internals or input data.

Plug-and-play API
Integrate in minutes. Standard endpoints, clear docs, no cryptography expertise required from developers.

Selective disclosure
Share only what's needed via the Midnight Network. Granular control over which data points are revealed.

Audit trail
Every inference is logged on-chain with privacy preserved. Tamper-proof records for compliance and trust.

Any model. Any data.
Fully private. Fully verifiable.

Any model

Upload and monetize any ML model without exposing weights or architecture

Any data

Run inference on sensitive data that never leaves the client's environment

Fully verifiable

Cryptographic proofs guarantee correct execution, no trust required

Turn ML into a trustless, monetizable primitive.